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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Course Code: CS2005                                                                                                                                                                     | Course Name: Database Systems |
| Instructor Name: Dr. Zulfiqar Ali Memon, Dr. Anam Qureshi, Ms. Abeer Gauher, Ms. Fizza Aqeel, Ms. Romasha Khurshid, Mr. Ali Fatmi, Mr. Shoaib Raza, Mr. Omer Qureshi, Ms. Fizza Mansoor |                               |
| Student Roll No:                                                                                                                                                                        | Section:                      |

**Instructions:**

- Return the question paper. Don't write anything on question paper, except your Roll # & Section #.
- Read each question completely before answering it. There are seven questions on three pages.
- In case of any ambiguity, you may make assumptions. But your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on answer sheet.

**Time:** 180 minutes.

**Max Marks:** 50

**Question 1 [7 marks] [CLO # 1]**

Assume that immediate update recovery is being used in a DBMS. A schedule at system crash is shown that includes four transactions.

Initial Values: A=30, B=120, C=30, D=20.

1. Design the log for the schedule shown Figure 01.
2. What are values after the crash for A, B, C and D.
3. Specify which transactions are rolled back and which (if any) are undone?

**Question 2 [7 marks] [CLO # 3]**

Consider the following two scenarios and write the algorithm for their deadlock prevention:

There are three transactions T1, T2 and T3 with the time stamp values as 10, 20, and 30 respectively.

1. T3 with the time stamp values as 10, 20, 30 respectively. If a transaction T1 requests a data item which is held by T2, then it is allowed to wait. If T3 requests a data item which is held by T2, then T3 will be rolled back (dies).
2. If a transaction T1 requests a data item which is held by T2 then the data item will be preempted from T2 and T2 is rolled back. Also, if T3 requests a data item which is held by T2 then T3 is allowed to wait since its time stamp is greater than that of transaction T2.

| T1                     | T2         | T3         | T4         |
|------------------------|------------|------------|------------|
| R(D)                   |            |            |            |
| D = D + 5              |            |            |            |
| W(D)                   |            |            |            |
| Commit                 |            |            |            |
|                        | R(B)       |            |            |
|                        | B = B + 60 |            |            |
|                        | W(B)       |            |            |
|                        |            |            | R(D)       |
|                        |            |            | D = D + 90 |
|                        |            |            | W(D)       |
|                        |            | R(C)       |            |
|                        |            | C = C + 10 |            |
|                        |            | W(C)       |            |
|                        |            |            | R(A)       |
|                        |            |            | A = A - 10 |
|                        |            |            | W(A)       |
|                        |            | Commit     |            |
|                        |            |            | R(A)       |
|                        |            |            | A = A - 5  |
|                        |            |            | W(A)       |
| *****System Crash***** |            |            |            |

**Figure 01: System Log**

**Question 3 [8 marks] [CLO # 1]**

Consider the following two transactions and some of their schedules:

**T1: r1(A),r1(B),w1(A),w1(B)**

**T2: r2(C),r2(B),w2(B),w2(C)**

S1: r1(A),r2(C),r1(B),r2(B),w1(A),w2(B),w1(B),w2(C),c1,c2

S2: r1(A),r1(B),w1(A),r2(C),r2(B),w2(B),w1(B),w2(C),c1,c2

S3: r1(A),r1(B),w1(A),w1(B),c1,r2(C),r2(B),w2(B),w2(C),c2

S4: r2(C),r2(B),w2(B),r1(A),r1(B),w1(A),w2(B),w2(C),c2,c1

S5: r1(A),r1(B), r2(C), w1(A),w1(B), r2(B),c1,w2(B),w2(C),c2

S6: r1(A),r1(B), r2(C), w1(A),w1(B), r2(B),w2(B),w2(C),c2,c1

Answer the following questions using transactions schedules given above:

1. Which schedule(s) is/are strict, serial schedule?
2. Is Schedule S3 recoverable?
3. Is schedule S4 cascadeless?
4. Which pair of schedules are conflict equivalent? Just give one such schedule?
5. Is schedule S5 a recoverable schedule?
6. Are schedule S1 and S5 result equivalent?
7. Is schedule S6 cascadeless?
8. Which pair of schedules are view equivalent? Just give one such schedule?

**Question 4 [8 marks] [CLO # 3]**

Cybersecurity is a highly competitive field that can be extremely rewarding for those with the right set of skills. Keeping in view of the market demand, FAST- NUCES have established the Cyber Security Department at Karachi Campus. The Table 01 below shows the details of Junior Year students registered in different courses:

Table 01: Registration Details of Junior Year Students

| Course_ID | Student_ID | Student_Name | Student_Age | Course_Name               | Classroom | Faculty_ID | Faculty_Name | Faculty_Contact |
|-----------|------------|--------------|-------------|---------------------------|-----------|------------|--------------|-----------------|
| CS2005    | 21k-4779   | Taha         | 21          | Database Systems          | R12       | 5773       | Ms. Tania    | 0300-2233446    |
|           | 21k-4504   | Rafay        | 22          |                           |           |            |              |                 |
|           | 21k-4518   | Sarim        | 22          |                           |           |            |              |                 |
| CS3001    | 21k-4701   | Amna         | 21          | Computer Networks         | E2        | 9100       | Mr. Ali      | 0333-2130523    |
|           | 21k-4732   | Bilal        | 22          |                           |           |            |              |                 |
|           | 21k-4702   | Shahid       | 21          |                           |           |            |              |                 |
| CS2001    | 21k-4167   | Abubakr      | 23          | Algorithms                | E5        | 5218       | Mr. Rahul    | 0345-4897524    |
|           | 21k-4478   | Saad         | 20          |                           |           |            |              |                 |
|           | 21k-4856   | Moiz         | 21          |                           |           |            |              |                 |
| CY3110    | 21k-5487   | Eman         | 22          | Malware Analysis          | E6        | 5485       | Mr. Asif     | 0334-5987412    |
|           | 21k-4476   | Riaz         | 23          |                           |           |            |              |                 |
| CY3105    | 21k-2559   | Moin         | 20          | Blockchain & Cryptography | E1        | 6582       | Mr. Zafar    | 0333-6987456    |
|           | 21k-4879   | Zain         | 23          |                           |           |            |              |                 |

1. Specify the insert, update, and delete anomalies, if any, for table 01.
2. Specify the primary key for table 01.
3. Convert the table up to 3NF by identifying the functional dependencies represented by the attributes. State any assumptions you make about the data shown in this table.
4. Design a relational Schema for your 3NF relations and show the referential integrity constraints.

**Question 5 [6 marks] [CLO # 2]**

Daraz.pk is an online shopping website that allows users to purchase different products. You are required to draw an ER Diagram that shows the relations between different entities along with the structural constraints.

Each customer has a unique customer\_id, name (composed of first and last name), address, and contact details. A customer can place multiple orders and each order has a unique order number, order date, and order amount. Each order has tracking details that allow customers to track their order. Tracking details include a unique tracking number, courier name, and courier contact number. An order can contain multiple products and each product has a unique product id, product name, product price, and product status ('Available', 'Out of Stock'). Each product has a product category with a unique category id and a category name. A supplier can supply multiple products and each supplier has a unique supplier id, supplier name and supplier number.

**Question 6 [7 marks] [CLO # 3]**

Assume the relational schema of a library management system where students can borrow the books and their records are maintained.

Note: DocId is Document ID, StId is Student ID, StName is Student Name, and AName is Author's name.

BOOKS (DocId, Title, Publisher, Year)

STUDENTS (StId, StName, Major, Age)

AUTHORS (AName, Address)

BORROWS (DocId, StId, Date)

HAS-WRITTEN (DocId, AName)

DESCRIPTION (DocId, Keyword)

Your task is to write the relational algebraic expressions of the following:

1. List the year and title of each book.
2. List all books published by McGraw-Hill before 1990.
3. List the name of students who are older than 30 and who are not studying CS
4. List the names of all students who have borrowed a book and who are CS majors.
5. Find the name of the youngest student.
6. Find the title of the oldest book.
7. Which books have both keywords 'database' and 'programming'?

**Question 7 [7 marks] [CLO # 3]**

Consider a relational schema of the salesman management system. Following are the relations and their attributes:

Salesman(Salesman\_ID, Name, City, Commission)

Customer(Customer\_ID, Customer\_Name, City, Grade, Salesman\_ID)

Order(Order\_Number, Purchased\_amount, Order\_date, Customer\_ID, Salesman\_ID)

Use the above schema to write the SQL queries of the following statements:

1. Find all orders generated by Karachi-based salesmen. Return ord\_no, purchased\_amt, ord\_date, customer\_id, and salesman\_id.
2. Determine the commission of the salesmen in Lahore. Return commission.
3. Find salesmen who had more than one customer. Return salesman\_id and name.
4. Find those customers whose grades are different from those living in Karachi. Return customer\_id, cust\_name, city, grade, and salesman\_id.
5. Calculate the total order amount generated by a salesmen. Salesmen should be from the cities where the customers reside. Return salesmen name, city, and total order amount.
6. Find those orders where the order amount exists between 500 and 2000. Return ord\_no, purchased\_amt, cust\_name, city.
7. Find the details of an order. Return ord\_no, ord\_date, purchased\_amt, Customer Name, grade, Salesman, and commission.

Good Luck!